Current GPU model — shared-reciprocal textured pipeline
N20K-SYS-0.1 — host, ordered execution, shared SDRAM and independent scanout

Host CPU / driver — outside simulated GPU execution time

Transform and homogeneous clipping
Projection / Q4 triangle setup / lighting
Stable bins for 300 logical 32×32 tiles

prepared commands and pixels

HostPassCompiler / CPU packet encoder
Persistent images + 320-byte prepared records
Tile table + ordered address lists
Upload new/changed textures + transient heap
Typical stream: WRITE_HEAP, SUBMIT_PASS,
then separate FENCE and PRESENT commands

serialized request bytes

Current device model — one common 100.8 MHz core / SDR clock (configured assumption)

Clocked host interface — no direct C++ pass injection

Shared 8-bit parallel bus
25.2 MHz derived transfer cadence
≤1 byte / 4 core edges, one direction
Setup 1 / hold 1 core cycle
Register select + valid/read/ready

accepted write byte

RX FIFO: 256 bytes
1 × SDPB 1024×16
Low byte per native word

registered read / capture

Byte parser / protocol checks
16-byte NGPU header, version 0.1
Length ≤1024, opcode and sequence
Partial packets never execute

write staged payload
publish only complete packet

3 × 1024-byte packet banks
3 × SDPB 1024×16
2 published command slots
INCLUDING the active front lease
+ 1 decoder-owned staging bank

front-bank lease

Synchronous payload read port
Executor holds front-bank ownership
until that command retires

fresh packet boundary/read selector
out-of-band controls
snapshot / capability byte
reply-byte read
latched request fields

Reply construction / serializer
1 × SDPB 1024×16
1 byte write / core edge
Separate synchronous read / capture
Reply: sequence + epoch + token

registered reply byte

TX reply FIFO: 1024 bytes
1 × SDPB 1024×16
Registered RAM + byte prefetch
Host pauses propagate backpressure

Sideband reads bypass command queue
Regs 1–63: coherent status snapshot
Regs 64–127: immutable capabilities
Status remains readable during faults

Out-of-band abort / reset
Abort: quiesce, drain issued memory,
flush queues, new epoch
Reset: restart initialization

cancel / drain / recover
established status, GET_CAPS / GET_STATUS reply

Ordered executor — exactly one command operation active; branches are alternatives

SystemModel executor FSM
Capture payload through packet RAM
Validate ranges / attachment ownership
Select one operation; retain its lease

opcode 5
opcode 3
opcode 4
address + length + data
completion
completion after swap

COPY_RECT / heap-image copy
Outside a render pass
Full clipped row, ≤640 RGB565 pixels
Reuse texture-cache bank as scratch
Row direction preserves overlap

WRITE_HEAP
Writer: one 32-bit pack + 4-bit mask
Source ≤1 byte / core edge
Uploads cannot write attachments

SUBMIT_PASS
One-record lookahead during current draw
One resident + one completed tile
One-edge bank-role exchange; no pixel copy
CLEAR export overlaps next tile; LOAD serialized
14 rendering banks +3 existing staging banks

opcode 7
opcode 6
opcode 8
reply published to TX
next complete command

PRESENT
Seal initialized writable target
Wait for eligible vertical blank
Swap front; then publish completion

destination writes acknowledged
write acknowledgements

FENCE
Execute after earlier stores complete
Publish host token / sequence / epoch

texture owner

READ_MEMORY
Ordered read, length 1–512 bytes
Uses timed shared reader
No dirty-tile inspection shortcut

final tile stores acknowledged
copy owner; invalidate tags

timed color copy
cache miss only
timed upload
fetch prepared work
attachment transfers

Command retirement
Release front packet bank
Advance ordered command stream

Resource alias, not another RAM
1 × SDPB 1024×16
Texture cache OR copy-row scratch
Mutually exclusive owners

timed read

Shared memory subsystem — six bounded client ports, one physical SDR data bus

Copy client
Source reads / masked destination writes

Texture client
External cache refill reads

Scanout client
Current front-buffer reads

Host client
Heap upload writes

Command client
Descriptor / table / reference / record
Diagnostic READ_MEMORY
Slot-read stream directly captures record fields

Tile client
Attachment boot writes
Color / depth loads and stores

Initialization and mandatory refresh
≥100 µs wait → PRECHARGE →
2 refreshes → MODE / timing waits
Refresh deadline ≤15.625 µs
≤1575 cycles at 100.8 MHz
Worst-case request / precharge guard

Per-client ownership / backpressure
1 request or unconsumed response slot
1–16 × 32-bit words, ≤64 bytes
2 × SDPB 1024×16 per client: 12 banks
Shared write-request / read-response data
Word-fill + synchronous read ports
Held response keeps its slot occupied

reserve service before deadline
queued request / response

Request arbitration
Scanout priority quota ≤4 requests
when other clients are waiting
Other five clients round-robin
One active request runs to completion

established vertical-blank pulse
swap only at vblank

grant / registered completion

Clocked SDRAM controller / native pins
Open rows / bank selection / ACTIVATE
PRECHARGE / CAS / write recovery
Read↔write turnaround / DQM masks
Sequential native BL8 (32-byte bursts)
Short / unaligned requests still pay BL8

request / response

per-edge observations

command / address / x32 DQ beats

Embedded SDR SDRAM: 8 MiB, x32
4 banks ×2048 rows ×256 columns
403.2 MB/s raw peak at 100.8 MHz
Word mapping: row11 | bank2 | col8
2 × RGB565 color + shared D24/U32
Heap: textures / records / descriptors / bins

Independent scanout — continues while the ordered executor renders or waits

Free-running display timing
H: 640 +16 front +96 sync +48 back
V: 480 +10 front +2 sync +33 back
Startup stays black until valid memory

pixel enable / H / V

Front / back ownership
Front is read-only to draws / copies
Old front becomes back after swap

current physical color ID

Front-buffer prefetch
1 outstanding tagged SDRAM read
Owned slot: synchronous 32-bit reads
One 32-bit hold → two RGB565 writes
Stale responses discarded at swap

one RGB565 write / edge; slot-read bubbles

RGB565 display FIFO: 2048 pixels
2 × SDPB 1024×16
Synchronous read → later capture
Smaller limits are stress settings

old registered output

Pixel consumer: core / 4 =25.2 MHz
640×480 active, 800×525 total
60 Hz, negative H/V sync
Underflow: black rest of frame
Resynchronize at vertical blank

captured completed scanout frame

Host-only observer
SDL / S-key screenshot show actual FIFO output
Raw dumps inspect completed framebuffer
Observer images are not GPU memory

Host-only cycle observer / throughput scoreboard
Distinct accept and retire IDs; stage / enable / occupancy
Common-clock CSV traces and cadence / latency histograms
Adds no GPU registers, queues or ports

Limits that matter
Pipelined fragments, record lookahead and completed CLEAR-tile export overlap
One active ordered command; heap uploads remain serialized
CPU wide validation checks are simulation assertions, not timed GPU units
39 native BSRAM =14 renderer +25 system I/O and staging
46 of48 native18×18 DSP equivalents; 64 textured contexts; bounded registers / logic ROM
Whole-system LUT / FF fit and achieved clock remain unproven
Solid teal: data / request-response. Dashed blue: control. Dotted purple: shared resource.
HDMI/TMDS signaling excluded.

Current GPU renderer — shared reciprocal and streaming fragments
N20K-HW-RENDER-0.3 · bounded contexts, backpressure and ordered retirement

1 · Primitive capture, setup and sample generation

Immutable prepared primitive
1 capture edge; bind texture
CPU clipping / triangle setup external
Drain before next primitive

Triangle initialization
6 products ×4 edges =24
ONE dedicated setup MULT18X18
Exact Δy + Δx plane-anchor walk

Triangle / 2D walker
Q4 centers and top-left coverage
Signed32 edges; signed64 planes
Rect / sprite grid; global line walk
At most one candidate / edge
Outside coverage: advance without token

covered candidate yes

2 · Separate untextured and textured paths

Covered candidate:
texture needed?

no

Untextured fast path
ONE pending fragment
Depth read → next-edge compare / commit
Can retire N while accepting N+1
Same-address depth forwarding

Textured context allocation
64-entry ordered ring / six-bit tag
Commit address + D24; RGB factor
Stop walker when credits unavailable

Early D24
Native read at acceptance
Compare on following edge
Reject before UV / sampling

Why no depth retest at retirement?
Each supported primitive visits a pixel once.
Primitives drain before another begins.
Future primitive overlap needs a hazard policy.

pass / disabled, textured triangle

pass / disabled, sprite

3 · Shared-reciprocal U/V pipeline — initiation interval ONE

Normalize q and signed U/V
Carry six-bit context tag
q raw domain [2^29, 2^37]
Sprites bypass this entire stage

Synchronous linear seed ROM
256 entries
35-bit base: THREE native banks
16-bit gradient: ONE native bank

Refine one reciprocal for U and V
Registered native18×18 partial products
40 DSP equivalents across full UV unit
No assumed free wide multiply

Bounded exact quotient correction
SignedQ16 round-to-nearest-even
Both U/V and error flags + context tag
26 enabled unit edges TOTAL
Parent captures on following edge

4 · Streaming texture hit path — initiation interval ONE, at most eight reservations

Address coordinates
Triangle: signedQ16 floor / clamp / repeat
Sprite: existing integer source X/Y
Preserve clipped source origin

Synchronous RGB-factor handoff
TWO SDPB banks, 64 ×RGB888
Read context tag → later capture
Two elastic coordinate/factor stages

Pitched word address
y × (pitch_low17 + pitch_high ×2^17) + x
TWO native registered18×18 DSPs
Separate address / tag registers

Tag and synchronous cache read
ONE SDPB1024×16 bank
64 direct-mapped lines ×16 texels
Native B read → later output capture
7 enabled unit edges on hit TOTAL
Parent captures on following edge

enabled writes

depth rejection

release oldest context

early native read

miss

line valid; replay held request

Miss service / replay
Freeze younger address and DSP stages
Older cache results drain in order
System: one owned Texture SDRAM slot
Sync32-bit reads → one held word
Two16-bit native writes per word
Pad final short-line tail without extra reads

captured with texture request

old registered texel

pop matching context

keep

5 · Raw key test and parallel RGB shading — initiation interval ONE

Raw RGB565
key match?

Matching texture-context FIFO
8 bounded register entries
Six-bit tag +24-bit RGB factor
Retain until ordered texture output

matching RGB factor

Replace or RGB modulation
THREE parallel registered18×18 DSPs
3 unit edges + parent capture
Exact rounded /255 shift-add
Sprites / replacement use factor255
through the same RGB stages

Store RGB565 result
ONE native context-result bank
Result +ready update on one edge
Scheduler sees OLD ready flags
Independent color/depth write controls

key rejection

next-edge retirement scheduler

6 · Ordered completion and attachment commit

Bounded ready/reason registers
64 ready bits +64 ×2-bit reasons
Commit / depth reject / key reject
A younger ready entry waits for older work

Retirement context ports
THREE commit-metadata SDPB banks
Tile address10 +D24, 64 entries
ONE RGB565 result bank, shared with shade
Synchronous read → separate capture

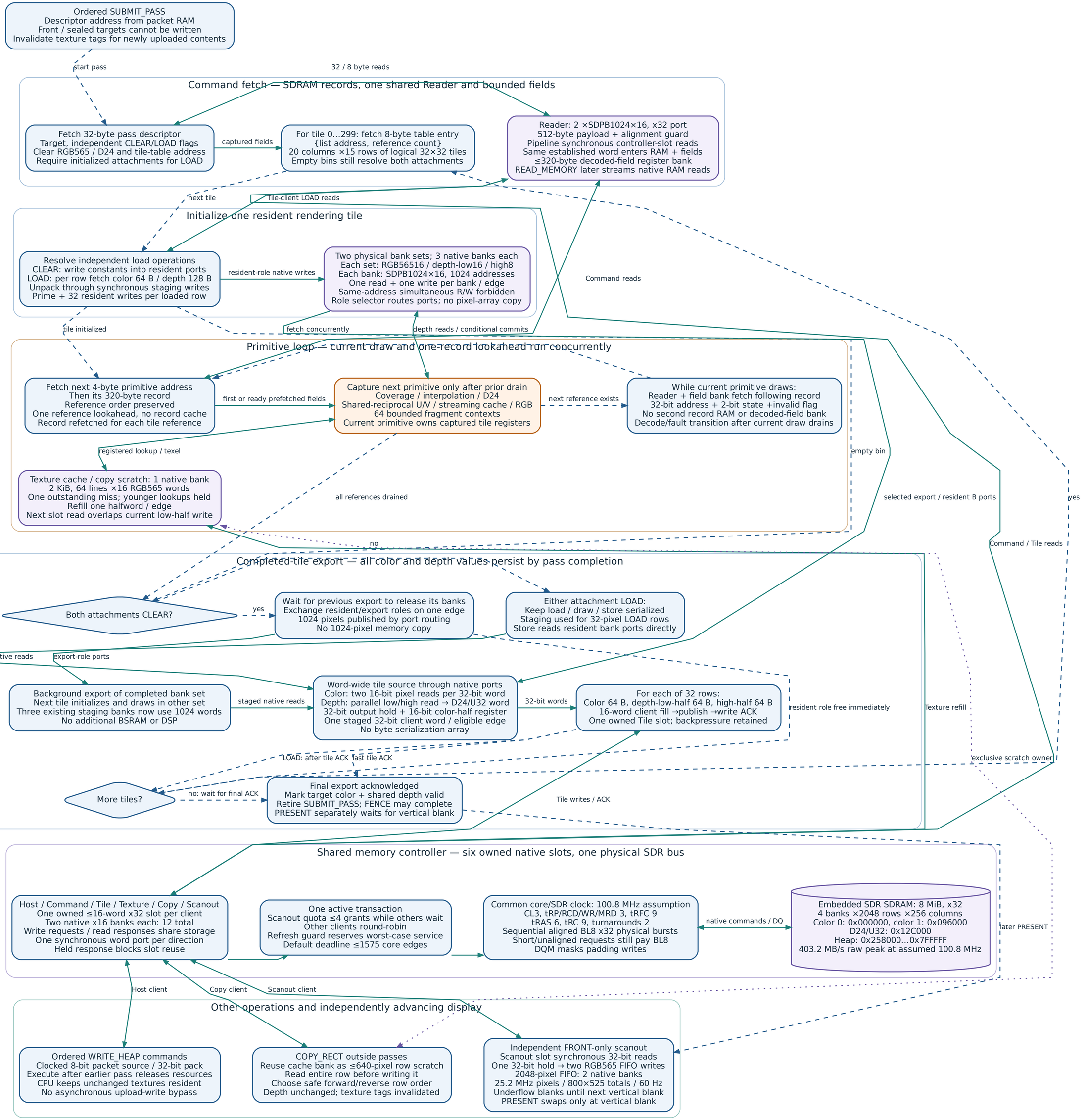
Retire oldest fragment once
Conditional RGB565 and D24 writes
Rejects write neither attachment
Release context credit
One retirement may overlap acceptance

Resident32×32 tile
THREE SDPB1024×16 banks
RGB565 / depth low16 / depth high8
Unused high-depth bits remain zero

SELECTED RESOURCES / CLOCK QUALIFICATION
Renderer:14 BSRAM =3 tile +1 cache/copy +6 context +4 seed ROM
System:39 of46 BSRAM =renderer14 +2 scanout +5 host +1 reply +12 client slots +2 reader +3 row
46 of48 native18×18 equivalents =1 setup +40 UV +2 texture address +3 RGB
The selected schedule permits overlap; misses/backpressure/context exhaustion cause bubbles.
Core/SDRAM100.8MHz is a configurable assumption. Whole-GPU fit and achieved timing are not established.
Old serial divider /14-edge texture-hit APIs remain standalone oracles, outside selected GPU execution.

HOST-ONLY OBSERVATION
Per-edge accepted and retired tokens; batch cancellation during reset/recovery
Walker phase is not exclusive utilization of concurrent arithmetic/cache/shading blocks
Hardware activity counters overlap: do not sum them into elapsed clocks
System last_pass_cycles includes record fetch, tile DMA, drawing and memory waits
Trace IDs, scoreboards and CSV/JSON create no FPGA state or GPU memory ports

Current GPU model — overlapped tile export and geometry lookahead
N20K-SYS-0.1 — exact native ports, ordered commands and shared SDRAM



Traffic / accounting
Each pass stores 1,843,200 useful color+depth bytes; LOAD of both adds 1,843,200 read bytes
Boot explicitly writes 2,457,600 attachment bytes; exclude boot when measuring steady-state submissions
39 native BSRAM = 14 renderer + 25 system; 46/48 native 18×18 DSP equivalents
One resident +one completed tile use two existing three-bank sets selected by a role mux
Eight 64-bit optimization counters; bounded lookahead/export controls; no hidden record payload storage
Fit and achieved clock unproven; model FPS is not measured FPGA performance
Host observers, CPU texture snapshots and diagnostics add no GPU memory ports
Solid teal: data. Dashed blue: control. Dotted purple: shared resource. HDMI signaling excluded.